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Candidate surname		Other names	
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Pearson Edexcel Award

Tuesday 23 June 2026

Afternoon (Time: 3 hours)

Paper reference **9811/01**

Advanced Extension Award

Mathematics *Worked Solutions*

You must have:
Mathematical Formulae and Statistical Tables
An insert for questions 5, 6 and 7 (enclosed)

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1)** or **(+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

$$y = \operatorname{cosec} x \quad x \in \mathbb{R} \quad x \neq n\pi \quad n \in \mathbb{Z}$$

where x is measured in radians.

By writing $\operatorname{cosec} x = \frac{1}{\sin x}$, use differentiation from first principles to prove that

$$\frac{dy}{dx} = -\operatorname{cosec} x \cot x$$

You may

- use without proof the formulae for $\sin(A \pm B)$
- assume that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$

You may **not** use without proof

- the chain rule
- the product rule
- the quotient rule

(6)

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\frac{1}{\sin(x+h)} - \frac{1}{\sin(x)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x) - \sin(x+h)}{h \sin(x+h) \sin(x)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x) - [\sin(x) \cos(h) + \cos(x) \sin(h)]}{h \sin(x+h) \sin(x)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x) [1 - \cos(h)]}{h \sin(x+h) \sin(x)} - \frac{\cos(x) \sin(h)}{h \sin(x+h) \sin(x)}$$

Since $\frac{\cos(h) - 1}{h} \rightarrow 0$ then $\frac{1 - \cos(h)}{h} \rightarrow 0$

so $\frac{\sin(x) [1 - \cos(h)]}{\sin(x+h) \sin(x) h} \rightarrow 0$

Since $\frac{\sin(h)}{h} \rightarrow 1$ $-\frac{\cos(x) \sin(h)}{\sin(x+h) \sin(x) h} \rightarrow -\frac{\cos(x)}{\sin(x+h) \sin(x)}$



Question 1 continued

$$\sin(x+h) \rightarrow \sin(x) \text{ as } h \rightarrow 0$$

$$\frac{dy}{dx} = - \frac{\cos(x)}{\sin(x) \sin(x)} = - \frac{1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)}$$

$$\frac{dy}{dx} = -\operatorname{cosec}(x) \cot(x)$$

Question 1 continued

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Question 1 continued

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(Total for Question 1 is 6 marks)

2.

$$f(x) = x^2 - 3x + 5 \quad x \leq k$$

where k is a constant.

Given that $f(x)$ has an inverse function,

(a) determine the maximum possible value of k .

(2)

For the value of k found in part (a),

(b) determine $f^{-1}(x)$, stating the domain of f^{-1}

(3)

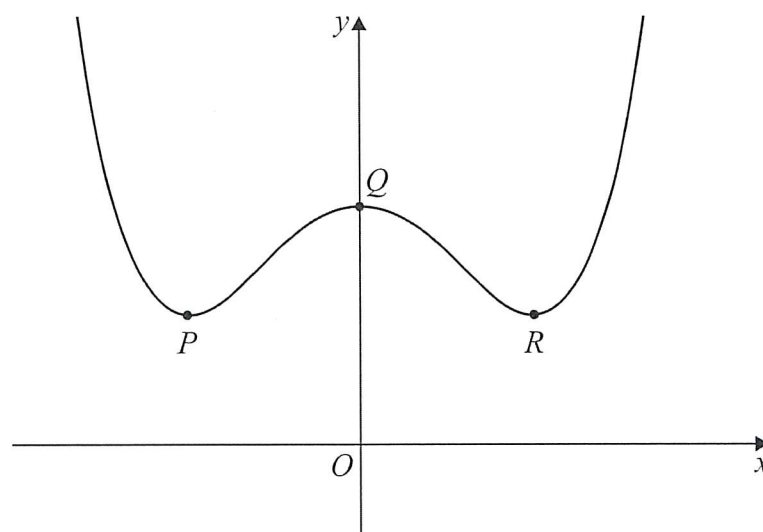


Figure 1

Figure 1 shows a sketch of the curve with equation $y = g(x)$, where

$$g(x) = x^4 - 3x^2 + 5$$

The points P , Q and R , also shown in Figure 1, are the turning points of the curve with equation $y = g(x)$.

Given that

$$h(x) = x^4 - 3x^2 + 5 \quad 0 \leq x \leq a$$

where a is the x coordinate of R ,

(c) use the answer to part (b), or otherwise, to determine $h^{-1}(x)$, stating the domain of h^{-1}

(2)

If $f(x)$ has an inverse then it must be a one to one function.

a) $f(x) = \left(x - \frac{3}{2}\right)^2 + \dots$ turning point occurs when $x = \frac{3}{2}$

$k = 3/2$



Question 2 continued

$$b) y = \left(x - \frac{3}{2}\right)^2 - \frac{9}{4} + 5$$

$$y = \left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$$

$$\pm \sqrt{y - \frac{11}{4}} = x - \frac{3}{2}$$

$$x = \frac{3}{2} - \sqrt{y - \frac{11}{4}}$$

Since $x \leq 3/2$

cannot be $\frac{3}{2} + \sqrt{y - \frac{11}{4}}$

$$f^{-1}(x) = \frac{3}{2} - \sqrt{x - \frac{11}{4}}$$

$$x \geq \frac{11}{4}$$

Since $x - \frac{11}{4} \geq 0$

$$c) h(x) = f(x^2)$$

$$x^2 = \frac{3}{2} - \sqrt{y - \frac{11}{4}}$$

$$x = \sqrt{\frac{3}{2} - \sqrt{y - \frac{11}{4}}}$$

$$h^{-1}(x) = \sqrt{\frac{3}{2} - \sqrt{y - \frac{11}{4}}}$$

$h(x)$ is $g(x)$ between point Q and R

domain of $h^{-1}(x)$ = range of $h(x)$

at Q $h(x) = 0 - 0 + 5 = 5$

at R $h(x) = \frac{11}{4}$ from turning point in (b)

$$\frac{11}{4} \leq x \leq 5$$

Question 2 continued

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Question 2 continued

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(Total for Question 2 is 7 marks)



3. A discrete random variable X takes positive integer values only.

The probability distribution of X is given by

- $P(X=1) = \frac{1}{3}$

- $P(X=2) = a$

where a is a constant, $0 < a < 1$, and

- if k is an **odd** integer and $k > 1$, then $P(X=k) = \frac{1}{9} P(X=k-2)$
- if k is an **even** integer and $k > 2$, then $P(X=k) = \frac{1}{\sqrt{5}} P(X=k-2)$

This is illustrated in the table below for the first six values of x

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{3}$	a	$\frac{1}{27}$	$\frac{1}{\sqrt{5}}a$	$\frac{1}{243}$	$\frac{1}{5}a$

- (a) Determine, in terms of a where appropriate,

(i) $P(X=7)$

(ii) $P(X=8)$

(1)

- (b) Show that

$$a = \frac{5 - \sqrt{5}}{8}$$

(2)

- (c) Determine $P(1 < X \leq 8 \mid X \text{ is even})$, giving the answer as a rational number.

(3)

Three independent observations of X are made, denoted by X_1 , X_2 and X_3

The variable Y is defined as $Y = X_1 + X_2 + X_3$

- (d) Determine $P(Y < 5)$, giving the answer in simplest form.

(3)

$$\begin{aligned} \text{a) } P(X=7) &= \frac{1}{9} \quad P(X=5) = \frac{1}{9} \times \frac{1}{243} = \frac{1}{2187} \\ P(X=8) &= \frac{1}{\sqrt{5}} \quad P(X=6) = \frac{1}{\sqrt{5}} \times \frac{1}{5} a = \frac{a}{5\sqrt{5}} \end{aligned}$$



Question 3 continued

b) For odd integers $\frac{1}{3} \quad \frac{1}{27} \quad \frac{1}{243} \dots$
we have a

geometric sequence $a = \frac{1}{3} \quad r = \frac{1}{9}$

for even $a \quad \frac{1}{\sqrt{5}} a \quad \frac{1}{5} a$

1st term = a $r = \frac{1}{\sqrt{5}}$

Sum of probabilities = 1

$$S_{\infty}(\text{odd}) + S_{\infty}(\text{even}) = 1$$

$$\frac{\frac{1}{3}}{1 - \frac{1}{9}} + \frac{a}{1 - \frac{1}{\sqrt{5}}} = 1$$

$$\frac{\frac{3}{8}}{1 - \frac{1}{\sqrt{5}}} = 1$$

$$\frac{a}{1 - \frac{1}{\sqrt{5}}} = \frac{5}{8}$$

$$a = \frac{5}{8} \left(1 - \frac{1}{\sqrt{5}} \right)$$

$$a = \frac{5}{8} - \frac{5}{8\sqrt{5}}$$

$$a = \frac{5\sqrt{5} - 5}{8\sqrt{5}}$$

$$a = \frac{5 - \sqrt{5}}{8}$$

$$\begin{aligned} \text{c) } P(1 < X \leq 8 | X \text{ is even}) &= \frac{P(1 < X \leq 8 \cap X \text{ is even})}{P(X \text{ is even})} \\ &= \frac{P(2) + P(4) + P(6) + P(8)}{S_{\infty}(\text{even})} \\ &= \frac{a + a/\sqrt{5} + a/5 + a/5\sqrt{5}}{5/8} \end{aligned}$$

Question 3 continued

$$= \frac{a \left(1 + \frac{1}{\sqrt{5}} + \frac{1}{5} + \frac{1}{5\sqrt{5}} \right)}{5/8}$$

$$= \left(\frac{5 - \sqrt{5}}{8} \right) \left(1 + \frac{1}{\sqrt{5}} + \frac{1}{5} + \frac{1}{5\sqrt{5}} \right) \times \frac{8}{5}$$

$$= \left(5 + \frac{5}{\sqrt{5}} + 1 + \frac{1}{\sqrt{5}} - \sqrt{5} - 1 - \frac{\sqrt{5}}{5} - \frac{1}{5} \right) \times \frac{1}{5}$$

$$= \frac{25\sqrt{5} + 25 + 5\sqrt{5} + 5 - 25 - 5\sqrt{5} - 5 - \sqrt{5}}{5\sqrt{5}} \times \frac{1}{5}$$

$$= \frac{24\sqrt{5}}{5\sqrt{5}} \times \frac{1}{5} = \frac{24}{25}$$

d) The lowest outcome is $1 + 1 + 1$ $Y=3$

$$P(Y < 5) = P(3) + P(4)$$

$$P(Y=3) \quad 1, 1, 1 \quad \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}$$

$$P(Y=4) \quad \begin{array}{l} 1 \quad 1 \quad 2 \\ \text{or} \quad 1 \quad 2 \quad 1 \\ \text{or} \quad 2 \quad 1 \quad 1 \end{array} \quad \frac{1}{3} \times \frac{1}{3} \times a$$

$$P(Y < 5) = \left(\frac{1}{3} \right)^3 + \left(\frac{1}{3} \right)^2 a \times 3$$

3 possible combinations

$$= \frac{1}{27} + \frac{1}{3} \times \frac{5 - \sqrt{5}}{8}$$

$$= \frac{8}{216} + \frac{45 - 9\sqrt{5}}{216}$$

$$= \frac{53 - 9\sqrt{5}}{216}$$



Question 3 continued

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(Total for Question 3 is 9 marks)



4.

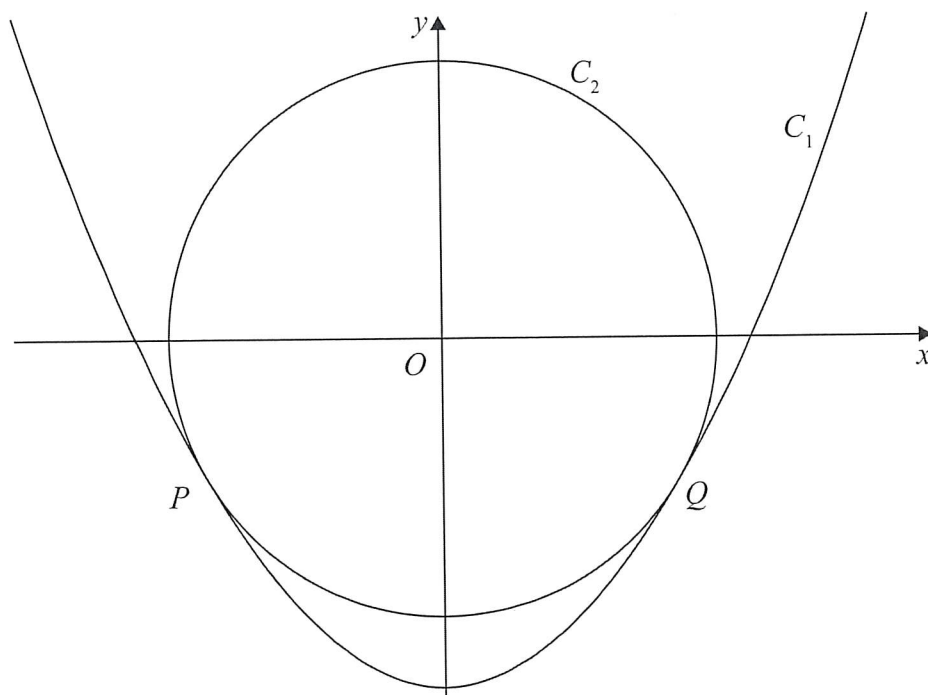


Figure 2

The curve C_1 , shown in Figure 2, has equation

$$y = Ax^2 + k$$

where A and k are constants with $A > 0$ and $k < 0$

The circle C_2 , also shown in Figure 2, has centre at the origin and radius r , where r is a positive constant.

Given that C_1 and C_2 touch at exactly two distinct points, P and Q ,

(a) (i) show that

$$k = -\frac{1}{4A} - Ar^2 \quad (4)$$

(ii) Hence determine the y coordinates of the points P and Q in terms of A only. (2)

(iii) Show that the x coordinates of the points P and Q are given by

$$x = \pm \sqrt{r^2 - \frac{1}{4A^2}} \quad (1)$$

(iv) Determine, in terms of A , the range of possible values of r . (1)



Question 4 continued

The curve C_3 has equation

$$y = Ax^2 + Bx + C$$

where B and C are constants and A is the constant in part (a).

The circle C_4 has centre (a, b) and radius r , where a and b are constants and r is the constant in part (a).

Given that C_3 and C_4 touch at exactly two distinct points, R and S ,

(b) (i) determine an equation for C_3

Give your answer in the form $y = f(x)$ in terms of A , a , b and r

(2)

(ii) determine the equations of all lines that are normal to both C_3 and C_4

Give the equations in terms of A , a , b and r where appropriate.

(4)

(+S1)

$$ai) C_2: x^2 + y^2 = r^2$$

$$C_1: \frac{y-k}{A} = x^2$$

$$\frac{y-k}{A} + y^2 = r^2$$

($\times A$)

$$y - k + Ay^2 = Ar^2$$

$$Ay^2 + y - k - Ar^2 = 0$$

$$b^2 - 4ac = 0$$

$$1 - 4A(-k - Ar^2) = 0$$

$$1 + 4Ak + 4A^2r^2 = 0$$

$$4Ak = -1 - 4A^2r^2$$

($\div 4A$)

$$k = \frac{-1}{4A} - Ar^2$$

$$aii) \text{ Substituting } k \text{ back into the quadratic in } y$$

$$Ay^2 + y - \left(\frac{-1}{4A} - Ar^2\right) - Ar^2 = 0$$

$$Ay^2 + y + \frac{1}{4A} = 0$$

$$y = \frac{-b \pm \sqrt{0}}{2a} \quad \text{Since we know } b^2 - 4ac = 0$$

$$y = \frac{-1}{2A}$$

Question 4 continued

aiii) $x^2 + y^2 = r^2$

$$x^2 + \left(-\frac{1}{2A}\right)^2 = r^2$$

$$x^2 = r^2 - \frac{1}{4A^2}$$

$$x = \pm \sqrt{r^2 - \frac{1}{4A^2}}$$

$$r^2 > \frac{1}{4A^2} \quad \text{since} \quad r^2 - \frac{1}{4A^2} > 0$$

$$r > \frac{1}{2A}$$

bi) C_4 is C_2 translated by $\begin{pmatrix} a \\ b \end{pmatrix}$

So C_3 is C_1 translated by $\begin{pmatrix} a \\ b \end{pmatrix}$

$$C_1: y = Ax^2 + k \quad C_3: y = A(x-a)^2 + k + b$$

$$\text{Since we know } k \quad C_3: y = A(x-a)^2 - \frac{1}{4A} - Ar^2 + b$$

bii) $x=0$ is normal to both C_1 and C_2 so
 $x=a$ is normal to both C_3 and C_4

To find the normals to C_3 and C_4 through
 R and S we can find the normals to
 C_1 and C_2 through P and Q then translate

$$\text{grad } OQ = \frac{-1/2A}{\sqrt{r^2 - \frac{1}{4A^2}}} = \frac{-1/2A}{\sqrt{\frac{1}{4A^2}} \sqrt{4Ar^2 - 1}} = \frac{-1}{\sqrt{4Ar^2 - 1}}$$

$$\text{grad } OP = -\text{grad } OQ = \frac{1}{\sqrt{4Ar^2 - 1}}$$



Question 4 continued

Other eq's of normal to C_3 and C_4 are

$$y = \frac{-1}{\sqrt{4A^2r^2 - 1}}(x - a) + b$$

$$y = \frac{1}{\sqrt{4A^2r^2 - 1}}(x - a) + b$$

Question 4 continued

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5.

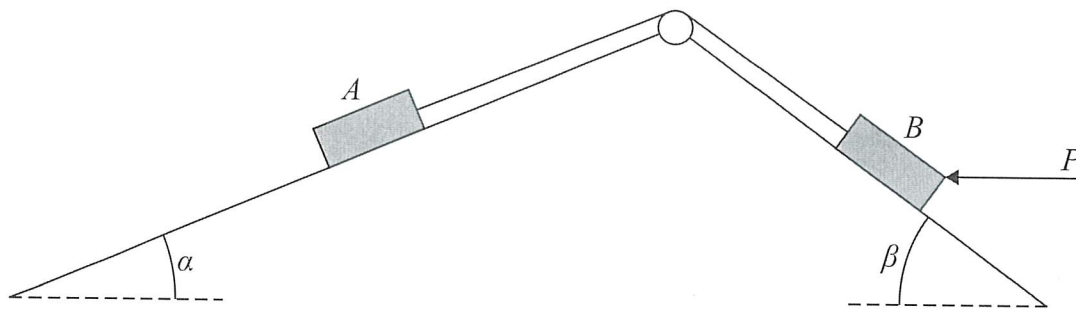


Figure 3

Figure 3 shows a block A with mass $26m$ and a block B with mass $75m$

Block A lies on a rough plane inclined at an angle α to the horizontal.

Block B lies on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth fixed light pulley.

The pulley lies on the line where the two inclined planes meet, with A , B and the pulley all lying in the same vertical plane.

The part of the string between A and the pulley lies along a line of greatest slope of the plane of angle α

The part of the string between B and the pulley lies along a line of greatest slope of the plane of angle β

The two blocks, A and B , are modelled as particles.

- $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{3}{4}$
- the coefficient of friction between block A and its plane is 0.5
- the coefficient of friction between block B and its plane is 0.2

A horizontal force of magnitude P acts on block B . The line of action of P lies in the same vertical plane as A , B and the pulley. The tension in the string is T .

When $k_1 mg \leq P \leq k_2 mg$, the system is at rest in equilibrium with the string taut.

- (a) Determine the exact values of k_1 and k_2 (9)

Assuming that the system remains in equilibrium,

- (b) determine, in simplest form in terms of T , the reaction force acting on the pulley. (3)

Question 5 continued

Without further calculation,

(c) explain the effect of decreasing P below $k_1 mg$ on

- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)

Without further calculation,

(d) explain the effect of increasing P above $k_2 mg$ on

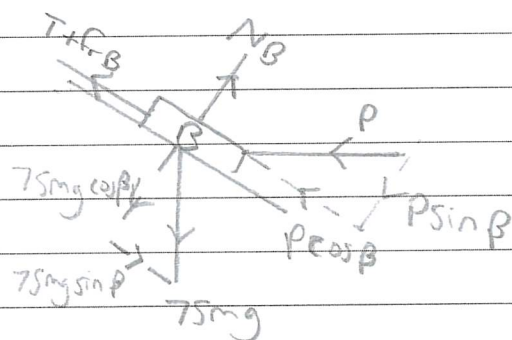
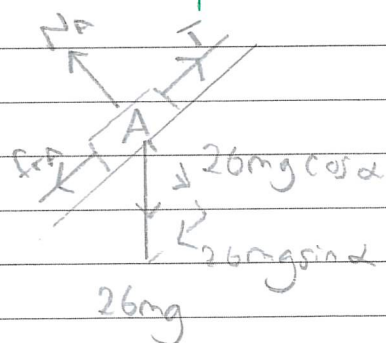
- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)
(+S2)

$$\sin \alpha = \frac{5}{13} \quad \cos \alpha = \frac{12}{13} \quad \sin \beta = \frac{3}{5} \quad \cos \beta = \frac{4}{5}$$

a) When $P = k_1 mg$ B is about to move down the plane



A perpendicular: $N_A = 26mg \cos \alpha = 24mg$

parallel: $T - f_A - 26mg \sin \alpha = 0$

$T = 10mg + f_A = 10mg + \frac{1}{2} N_A = 22mg$

B perpendicular: $N_B = 75mg \cos \beta + P \sin \beta = 60mg + \frac{3}{5} P$

parallel: $75mg \sin \beta - T - f_B - P \cos \beta = 0$

$T = 45mg - \frac{1}{5} N_B - \frac{4}{5} P$



Question 5 continued

$$22mg = 45mg - \frac{1}{5} \left(60mg + \frac{3}{5} P \right) - \frac{4}{5} P$$

$$\frac{23}{25} P = 11mg$$

$$P = \frac{275mg}{23}$$

$$k_1 = \frac{275}{23}$$

When $P = k_2 mg$ B is about to move up the plane.
As such, $T = 0$ and f_{rB} will reverse direction to oppose motion.

B perpendicular : $N_B = 60mg + \frac{3}{5} P$ as before

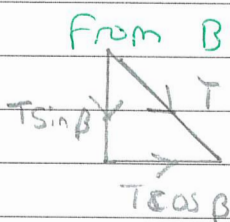
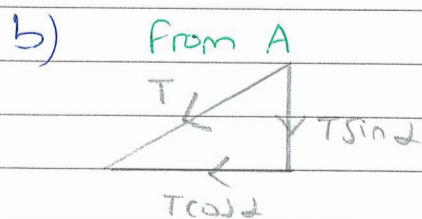
parallel : $P \cos \beta - f_{rB} - 75mg \sin \beta = 0$ since $a = 0$

$$\frac{4}{5} P = \frac{1}{5} \left(60mg + \frac{3}{5} P \right) + 45mg$$

$$\frac{17}{25} P = 57mg$$

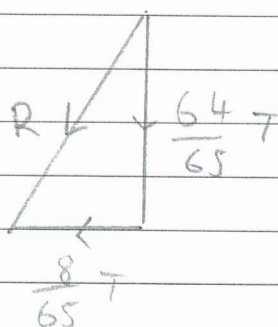
$$P = \frac{1425mg}{17}$$

$$k_2 = \frac{1425}{17}$$



$$\text{total vertical force} = \frac{5}{13} T + \frac{3}{5} T = \frac{25 + 39}{65} T = \frac{64}{65} T$$

$$\text{total horizontal force} = \frac{12}{13} T - \frac{4}{5} T = \frac{60 - 52}{65} T = \frac{8}{65} T$$



$$R^2 = \left(\frac{64}{65} T \right)^2 + \left(\frac{8}{65} T \right)^2$$

$$R^2 = \left(\frac{64}{65} T \right)^2 + \frac{64}{65^2} T^2$$

Question 5 continued

$$R^2 = \frac{64}{65} T^2 \left(\frac{64}{65} + \frac{1}{65} \right)$$

$$R^2 = \frac{64 T^2}{65} = \frac{65}{65} = 1$$

$$R = \frac{8T}{\sqrt{65}} = \frac{8\sqrt{65} T}{65}$$

ci) B would move down the plane as friction is already at its maximum so reducing P would mean the total resistive force is less than the weight component of B parallel to the plane.

cii) A is attached to B by a taut string so as B moves down A moves up the plane

ciii) The tension would increase so R would increase.

di) B would move up the plane as friction is already at its maximum so cannot increase to maintain equilibrium. $P \cos \beta > f_B + T \sin \beta$

dii) Since max f_A is greater than the weight component of A parallel to the plane, A would remain stationary.

diii) The string would no longer be taut so $T=0$ hence $R=0$.



Question 5 continued

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(Total for Question 5 is 20 marks)

6. Given that

$$y = \arctan x$$

(a) prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad (2)$$

Given that f is a function such that the integrals exist,

(b) (i) use the substitution $u = a + b - x$ to show that

$$\int_a^b f(x) \, dx = \int_a^b f(a + b - x) \, dx$$

where a and b are constants

(ii) determine the value of the constant λ such that

$$\int_a^b f(x) \, dx = \lambda \int_a^b (f(x) + f(a + b - x)) \, dx \quad (3)$$

Given that $I = \int_{-\pi}^{\pi} \frac{x(1 + \sin x)}{1 + \cos^2 x} \, dx$

(c) use detailed reasoning to show that

$$I = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx \quad (2)$$

Given that

$$g(x) = \frac{x \sin x}{1 + \cos^2 x}$$

(d) show that

$$g(x) = g(-x) \quad (1)$$

(e) Using the results from parts (a), (b), (c) and (d), determine the value of I

(8)
(+S2)



Question 6 continued

a) $\tan y = x$

$$\sec^2 y = \frac{dx}{dy}$$

$$1 + x^2 = \frac{dx}{dy}$$

$$\frac{1}{1+x^2} = \frac{dy}{dx}$$

$$\begin{aligned} 1 + \tan^2 y &= \sec^2 y \\ 1 + x^2 &= \sec^2 y \end{aligned}$$

bi) $u = a + b - x$

$$x = a + b - u$$

$$\frac{du}{dx} = -1 \Rightarrow -du = dx$$

When $x = a$ $u = a + b - a = b$

When $x = b$ $u = a + b - b = a$

$$\int_a^b f(x) dx = \int_b^a f(a+b-u) \cdot -du = \int_a^b f(a+b-u) du$$

Since u is a dummy variable $= \int_a^b f(a+b-x) dx$

bii) $\int_a^b f(x) dx = \lambda \int_a^b (f(x) + f(a+b-x)) dx$

$$= \lambda \left[\int_a^b f(x) dx + \int_a^b f(x) dx \right]$$

$$= 2\lambda \int_a^b f(x) dx \Rightarrow \lambda = \frac{1}{2}$$

c) Using bii

$$\int_{-\pi}^{\pi} \frac{x(1+\sin x)}{1+\cos^2 x} dx = \frac{1}{2} \int_{-\pi}^{\pi} \left[\frac{x(1+\sin x)}{1+\cos^2 x} + \frac{(\pi - \pi - x)(1+\sin(\pi - \pi - x))}{1+\cos^2(\pi - \pi - x)} \right] dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \left[\frac{x(1+\sin x)}{1+\cos^2 x} - \frac{x(1+\sin(-x))}{1+\cos^2(-x)} \right] dx$$

$\sin(-x) = -\sin(x)$ $\cos(-x) = \cos(x)$

Question 6 continued

$$= \frac{1}{2} \int_{-\pi}^{\pi} \frac{x + \sin x - x + \sin x}{1 + \cos^2 x} dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \frac{2 \sin x}{1 + \cos^2 x} dx = \int_{-\pi}^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$d) \quad g(-x) = \frac{-x \sin(-x)}{1 + \cos^2(-x)} = \frac{-x \cdot -\sin x}{1 + \cos^2 x} = \frac{x \sin x}{1 + \cos^2 x} = g(x)$$

$$e) \quad I = \int_{-\pi}^{\pi} g(x) dx$$

Since $g(-x) = g(x)$ g is an even function

$$\text{So } \int_{-\pi}^0 g(x) dx = \int_0^{\pi} g(x) dx$$

$$I = 2 \int_0^{\pi} g(x) dx$$

$$= \int_0^{\pi} g(x) dx + \int_0^{\pi} g(a+b-x) dx \quad \text{from b.c}$$

$$= \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx + \int_0^{\pi} \frac{(\pi + 0 - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$$\sin(\pi - x) = \sin(x)$$

$$\cos(\pi - x) = -\cos(x)$$

$$\cos^2(\pi - x) = (-\cos(x))^2 = \cos^2(x)$$

$$= \int_0^{\pi} \left[\frac{x \sin x}{1 + \cos^2 x} + \frac{\pi \sin x}{1 + \cos^2 x} - \frac{x \sin x}{1 + \cos^2 x} \right] dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx$$

$$\text{Let } u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$= \int_1^{-1} \frac{-\pi}{1 + u^2} du$$

$$\frac{du}{-\sin x} = dx \quad -du = \sin x dx$$

$$\text{When } x = 0 \quad u = 1$$

$$x = \pi \quad u = -1$$



Question 6 continued

$$= -\pi \int_1^{-1} \frac{1}{1+u^2} du$$

$$= -\pi [\arctan u]_1^{-1}$$

Using (a)

$$= -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= -\pi \times -\frac{\pi}{2} = \frac{\pi^2}{2}$$

Question 6 continued

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Question 6 continued

(Total for Question 6 is 18 marks)

7. $y = x^x \quad x > 0$

By taking logarithms of both sides, or otherwise,

- (a) determine $\frac{dy}{dx}$, giving the answer in simplest form in terms of x only. (3)

Given that, for $x > 0$

$$f(x) = (x^x)^x$$

$$g(x) = x^{(x^x)}$$

- (b) determine the exact coordinates of the turning point of $y = f(x)$ (5)
- (c) Show that $g'(x)$ can be written in the form

$$g'(x) = P(x)x^{Q(x)}$$

where $P(x) = x(\ln x)^2 + x \ln x + 1$ and $Q(x)$ is a function of x to be found. (4)

- (d) Solve $P(x) = 1$ (2)

Given that $y = P(x)$ has a stationary point at $x = \alpha$ and a stationary point at $x = \beta$, where $\alpha < \beta$,

- (e) determine the value of α and the value of β (3)
- (f) Hence show that $P(x)$ has a minimum at $x = \beta$ (2)

Using the answers to parts (c), (d), (e) and (f),

- (g) show that $g(x)$ is an increasing function for $x > 0$ (4)
(+S2)

$$\begin{aligned} \text{a)} \quad \ln y &= \ln x^x = x \ln x & U &= x & V &= \ln x \\ \frac{1}{y} \frac{dy}{dx} &= \ln x + \frac{x}{x} & U' &= 1 & V' &= \frac{1}{x} \\ \frac{dy}{dx} &= y (\ln x + 1) = x^x (\ln x + 1) \end{aligned}$$



Question 7 continued

b) $y = (x^x)^x$

$$\ln y = x \ln x^x = x^2 \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = 2x \ln x + \frac{x^1}{x}$$

$$\frac{dy}{dx} = y (2x \ln x + x) = 0$$

$$(x^x)^x \neq 0$$

$$2x \ln x + x = 0$$

$$2 \ln x + 1 = 0$$

$$\ln x = -\frac{1}{2}$$

$$U = x^x$$

$$U' = 2x$$

$$V = \ln x$$

$$V' = \frac{1}{x}$$

$$y = (x^x)^x = x^{x^2} = \frac{x = e^{-1/2}}{(e^{-1/2})^{(e^{-1/2})^2}} = (e^{-1/2})^{e^{-1}} = e^{-\frac{1}{2e}}$$

$$\text{T.P.} = (e^{-1/2}, e^{-1/2e})$$

c) $y = x^{(x^x)}$

$$\ln y = \ln x^{(x^x)} = x^x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = x^x (\ln x + 1) \ln x + \frac{x^x}{x}$$

$$\frac{dy}{dx} = x^{(x^x)} \left[x^x (\ln^2 x + \ln x) + \frac{x^x}{x} \right]$$

$$= x^{x^x} \cdot \frac{x^x}{x} \left[x (\ln^2 x + \ln x) + 1 \right]$$

$$= x^{x^x} \cdot x^{x-1} \left[x \ln^2 x + x \ln x + 1 \right]$$

$$= x^{x^x+x-1} \left[x \ln^2 x + x \ln x + 1 \right]$$

$$U = x^x$$

$$U' = x^x (\ln x + 1)$$

from (a)

$$V = \ln x$$

$$V' = \frac{1}{x}$$

d) $x \ln^2 x + x \ln x + 1 = 1$

$$x \ln x (\ln x + 1) = 0$$

$$x \neq 0 \quad \ln x = 0$$

$$x = e^0 = 1$$

$$\ln x + 1 = 0$$

$$\ln x = -1$$

$$x = e^{-1}$$

Question 7 continued

$$u = x \quad v = (\ln(x))^2$$

$$u' = 1 \quad v' = 2 \ln x \cdot \frac{1}{x}$$

$$u = x \quad v = \ln x$$

$$u' = 1 \quad v' = \frac{1}{x}$$

e) $P'(x) = 0$

$$P(x) = x \ln^2 x + x \ln x + 1$$

$$P'(x) = \ln^2 x + \frac{2x \ln x}{x} + \ln x + \frac{x}{x}$$

$$\ln^2(x) + 2 \ln x + \ln x + 1 = 0$$

$$\ln^2(x) + 3 \ln x + 1 = 0$$

$$\ln x = \frac{-3 \pm \sqrt{9 - 4 \times 1 \times 1}}{2} = \frac{-3 \pm \sqrt{5}}{2}$$

$$x = e^{\frac{-3 \pm \sqrt{5}}{2}}$$

$$\alpha = e^{\frac{-3 - \sqrt{5}}{2}}$$

$$\beta = e^{\frac{-3 + \sqrt{5}}{2}}$$

Since $\beta > \alpha$

f) $P''(x) = 2 \ln(x) \cdot \frac{1}{x} + \frac{3}{x}$

at β $P''(\beta) = \frac{2 \left(\frac{-3 + \sqrt{5}}{2} \right)}{e^{\frac{-3 + \sqrt{5}}{2}}} + \frac{3}{e^{\frac{-3 + \sqrt{5}}{2}}}$

$$= \frac{\sqrt{5}}{e^{\frac{-3 + \sqrt{5}}{2}}}$$

Since $e^{\frac{-3 + \sqrt{5}}{2}} > 0$ $P''(\beta) > 0$ hence minimum

g) If $g(x)$ is an increasing function for $x > 0$
then $g'(x) > 0$ for $x > 0$

$$g'(x) = P(x) x^{Q(x)}$$

Since $x > 0$ x raised to any power will be positive.
Hence $x^{Q(x)} > 0$

We know $P(x)$ has a minimum at $x = \beta = e^{\frac{-3 + \sqrt{5}}{2}}$
from a) we know $P(x)$ crosses $y=1$ at two points $x = 1(e^0)$ and $x = e^{-1}$

Since $\sqrt{5} < 3$ $e^{\frac{-3 + \sqrt{5}}{2}} < e^0$

Since $\sqrt{5} > -1$ $e^{\frac{-3 + \sqrt{5}}{2}} > e^{-1}$



Question 7 continued

Since $P(\beta)$ is a minimum and $P(x)$ crosses $y=1$ either side of $x=\beta$ only once we can deduce $P(x) \geq P(\beta)$ for $x > 0$

$$P(\beta) = \beta \ln \beta (\ln \beta + 1) + 1$$

$$= e^{\frac{-3+\sqrt{5}}{2}} \left(\frac{-3+\sqrt{5}}{2} \right) \left(\frac{-3+\sqrt{5}}{2} + 1 \right) + 1$$

$$= e^{\frac{-3+\sqrt{5}}{2}} \left(\frac{-3+\sqrt{5}}{2} \right) \left(\frac{-1+\sqrt{5}}{2} \right) + 1$$

$$= e^{\frac{-3+\sqrt{5}}{2}} \left(\frac{8-4\sqrt{5}}{4} \right) + 1$$

$$= e^{\frac{-3+\sqrt{5}}{2}} (2 - \sqrt{5}) + 1$$

$$e^{\frac{-3+\sqrt{5}}{2}} > 0 \quad \text{and from earlier}$$

$$e^{\frac{-3+\sqrt{5}}{2}} < e^0 = 1$$

$$-1 < 2 - \sqrt{5} < 0 \quad \text{since } 2 < \sqrt{5} < 3$$

$$\text{So } -1 < e^{\frac{-3+\sqrt{5}}{2}} (2 - \sqrt{5}) < 0$$

$$\text{and } 0 < e^{\frac{-3+\sqrt{5}}{2}} (2 - \sqrt{5}) + 1 < 1$$

Hence $P(\beta) > 0$ and $P(x) \geq P(\beta)$ so

$$P(x) > 0 \quad \text{for } x > 0$$

As $P(x) > 0$ and $x^{g(x)} > 0$ for $x > 0$

$$g'(x) > 0 \quad \text{for } x > 0$$

thus $g(x)$ is an increasing function for $x > 0$

Question 7 continued

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